

Student
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Total Points
3.5 / 4 pts

Question 1

Q11.5 / 2 pts

✓ + 0.5 pts $V_0 = \text{span}\{I[k-1/2, k+1/2] \text{ for integral } k\}$

✓ + 0.5 pts $k = 0$ gives us f i.e. f belongs to V_0

✓ + 0.5 pts $V_1 = \text{span}\{\sqrt{2}.I[k/2-1/4, k/2+1/4] \text{ for integral } k\}$

+ 0.5 pts f is not in V_1 i.e. linear combination can't reproduce f

+ 0 pts No/Wrong answer

Question 2

Q22 / 2 pts

✓ + 0.5 pts $H = -\sum(p \log p)$

✓ + 1.5 pts Correct answer: 5

+ 0 pts No/Wrong answer

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COL7(6)83 Quiz 7

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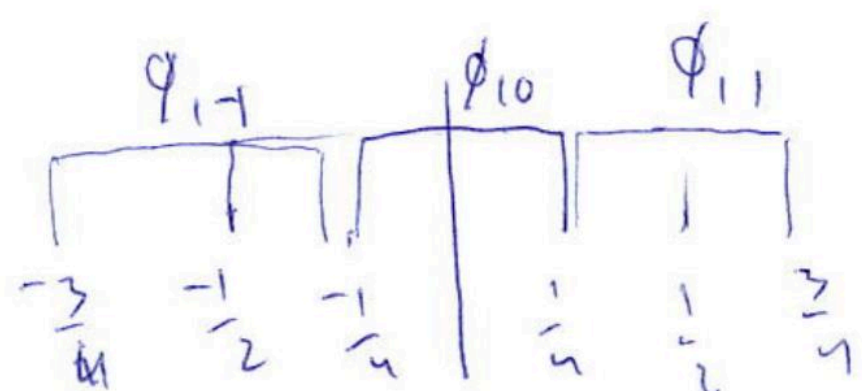
1. Explain why the unit box function on $[-\frac{1}{2}, \frac{1}{2}]$, i.e. $f(x) = \begin{cases} 1 & -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 0 & \text{otherwise,} \end{cases}$ cannot be used as the scaling function in a wavelet transform. (Hint: Show that $V_0 \not\subset V_1$.)

$$\phi_0 = f(x)$$

$$\phi_1 = \sqrt{2} f(2x)$$

there is no way to factorize ϕ_0 using ϕ_{1j}

here is a visual proof



$\phi_0 \in V_0$ but

ϕ_0 is not linear combo of ϕ_{1j}

$\phi_1 \notin V_1 \therefore V_0 \not\subset V_1$

2. Consider an image where half the pixels have intensity 0, and the other half have intensities uniformly distributed between 1 and 255. Thus, $P(0) = \frac{1}{2}$, $P(1) = P(2) = \dots = P(255) = \frac{1}{510}$. What is the entropy of this distribution? You may approximate $\log_2 \frac{1}{510} \approx \log_2 \frac{1}{512} = -9$.

$$H = - \sum p_i \log_2(p_i)$$

$$= - \left[\frac{1}{2} \cdot \log_2 \frac{1}{2} + \frac{255}{510} \cdot \log_2 \frac{1}{510} \right]$$

$$= + \frac{1}{2} \log_2 2 + \frac{255}{510} \cdot 9$$

$$= 5$$